Module 5: System Sequence Diagram & Activity Diagram
INFO C450 - System Design
Spring 2026
Titus Duncan II

Collaborative Media Management System (CollabSpace), Titus Duncan

1. Sequence Diagrams

1.1 System Sequence Diagram 1: Add Version (<> Upload Asset)

Actor: MemberAccount (Creator/Collaborator)

Objects: CollabSpace UI, AccessControl, AssetService, VersionService, Storage, Database

1. MemberAccount selects an asset / chooses a file in the CollabSpace UI.
2. CollabSpace UI requests upload authorization from AccessControl using memberId and projectId.
3. If access is granted, CollabSpace UI sends the file and metadata to AssetService to upload the asset.
4. AssetService stores the file in Storage.
5. AssetService saves the asset record (assetId, metadata, storage URL) in the Database.
6. CollabSpace UI requests VersionService to add a new version using the assetId and notes.
7. VersionService creates a new version record (versionId, assetId, notes, timestamp) in the Database.
8. VersionService sets the newly created version as the current version in the Database.
9. VersionService returns the created versionId to the CollabSpace UI.
10. CollabSpace UI displays a success confirmation to MemberAccount.

Alternatives:

- If unauthorized, then the UI will show “Access denied.”
- If upload fails, then the UI shows “Upload error.”

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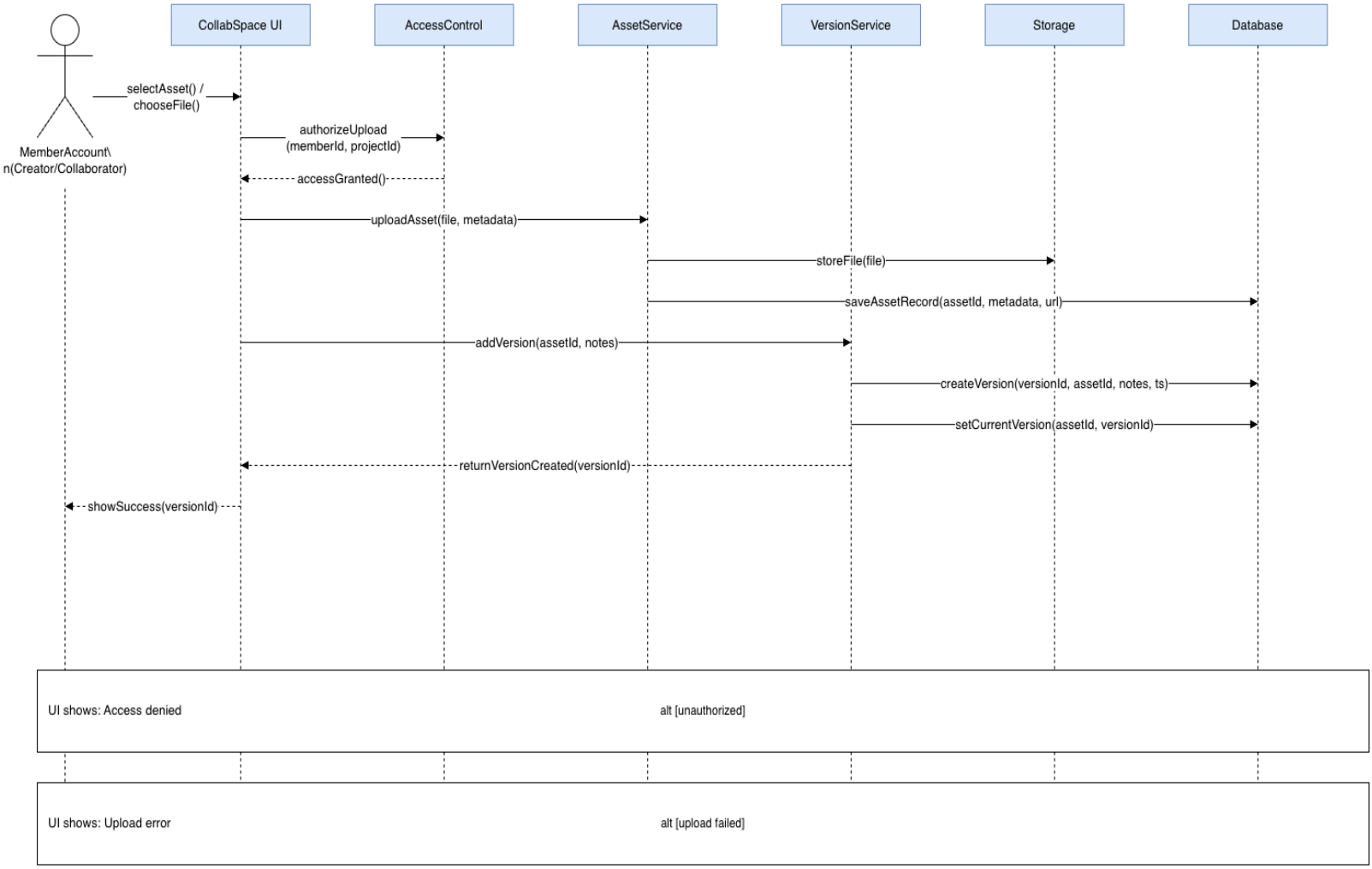
Sequence Diagram 1: Add Version



1.2 System Sequence Diagram 2: Approve Version

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Actor: Client (GuestViewer/ClientAccount)

Objects: CollabSpace UI, ControlledAccess, AccessControl, VersionService, NotificationService, Database

1. Client opens the presentation page using linkId in the CollabSpace UI.
2. CollabSpace UI asks ControlledAccess to validate the link.
3. ControlledAccess requests AccessControl to authorize the viewer (guest token or account) for the page.
4. If access is granted, AccessControl returns accessGranted() to the CollabSpace UI.
5. Client selects a version to review.
6. Client submits an approval with a note.
7. CollabSpace UI sends the approval request to VersionService.
8. VersionService creates an approval record in the Database.
9. VersionService updates the version status to APPROVED in the Database.
10. VersionService requests NotificationService to notify the Creator.
11. NotificationService returns confirmation that the notification is queued.
12. VersionService returns approval confirmation (approvalId) to the CollabSpace UI.
13. CollabSpace UI shows approval success to the Client.

Alternatives:

- If link is invalid or expired, then the UI will show “Link expired / invalid.”
- If access is denied, then the UI will show “Access denied.”
- If version is already approved or locked, then the UI will show “Cannot approve: status unchanged.”

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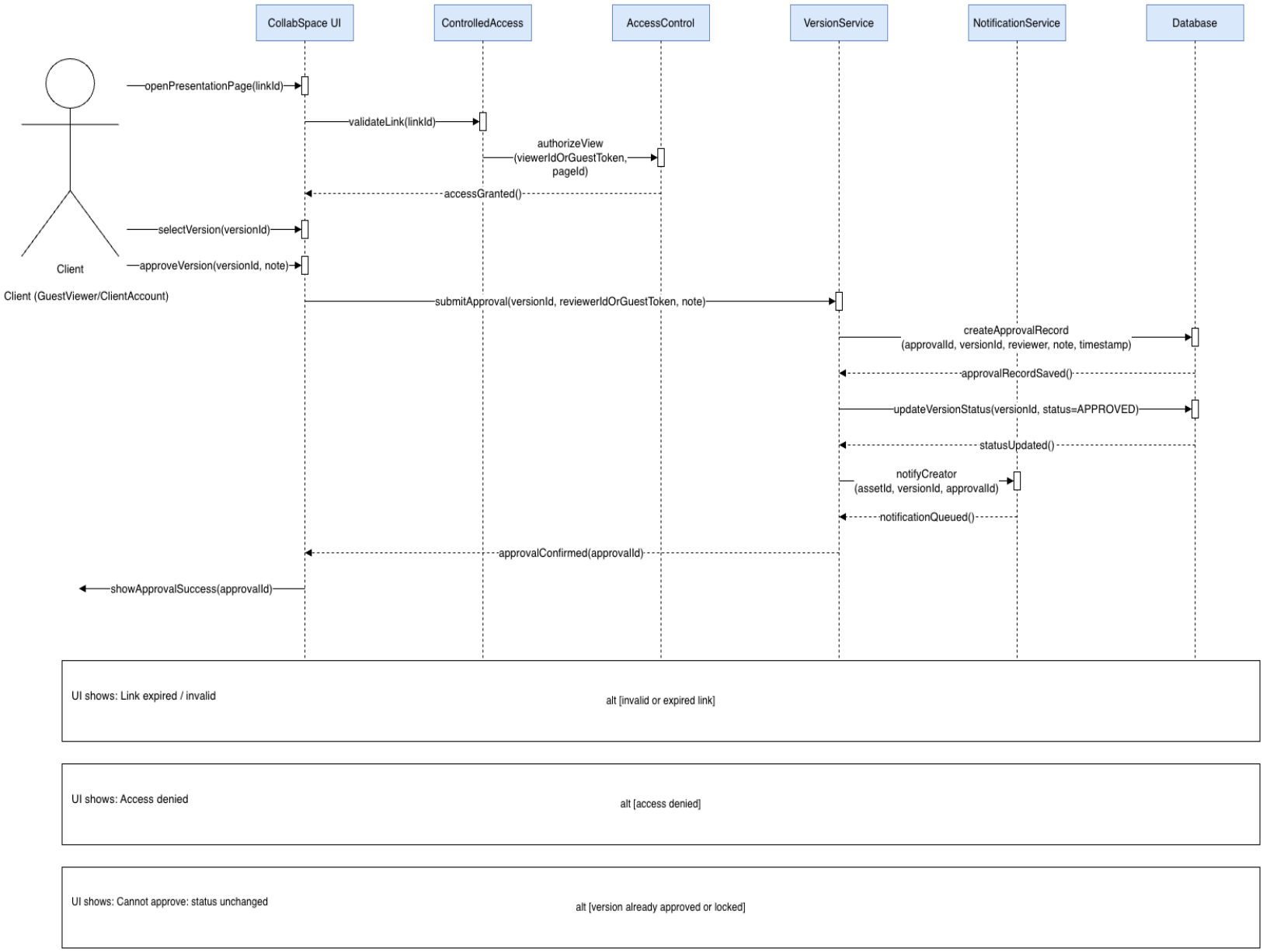
Sequence Diagram 2: Approve Version



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2. Activity Diagrams

2.1 Activity Diagram 1: Add Version (includes Upload Asset)

Initial State: MemberAccount starts adding a version by selecting a file.

Final State: Version is created successfully OR the process ends due to access denial or upload failure.

1. MemberAccount selects an asset / chooses a file.
2. System authorizes the upload request.
3. If not authorized, UI shows “Access denied” and the process ends.
4. If authorized, the system uploads the asset and stores it.
5. If upload fails, UI shows “Upload error” and the process ends.
6. If upload succeeds, the system saves the asset record.
7. The system adds a version and creates a version record.
8. The system sets the new version as the current version.
9. UI shows “Version created” and the process ends.

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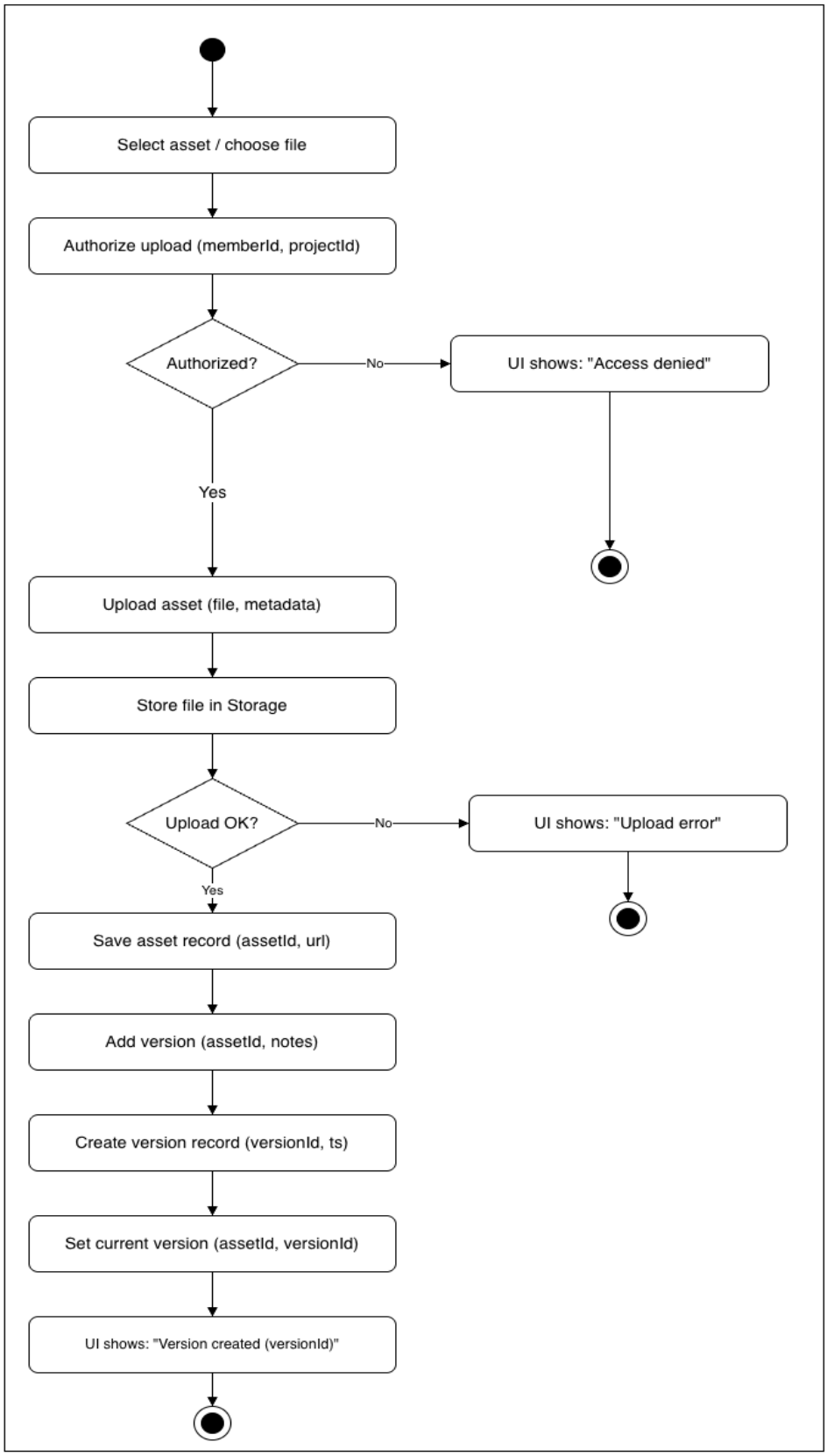
Activity Diagram: Add Version (includes Upload Asset)



2.2 Activity Diagram 2: Approve Version

Initial State: Client opens the presentation page from a link.

Final State: Approval is submitted successfully OR the process ends due to invalid link, access denial, or locked status.

1. Client opens the presentation page (linkId).
2. System validates the link.
3. If link is invalid, UI shows “Link expired / invalid” and the process ends.
4. If valid, system authorizes viewing (guest token / account).
5. If access is denied, UI shows “Access denied” and the process ends.
6. If access is granted, client selects a version.
7. Client submits an approval note.
8. System checks whether the version is locked or already approved.
9. If locked/already approved, UI shows “Cannot approve: status unchanged” and the process ends.
10. If allowed, system creates an approval record.
11. System updates the version status to APPROVED.
12. System notifies the creator.
13. UI shows “Approval success” and the process ends.

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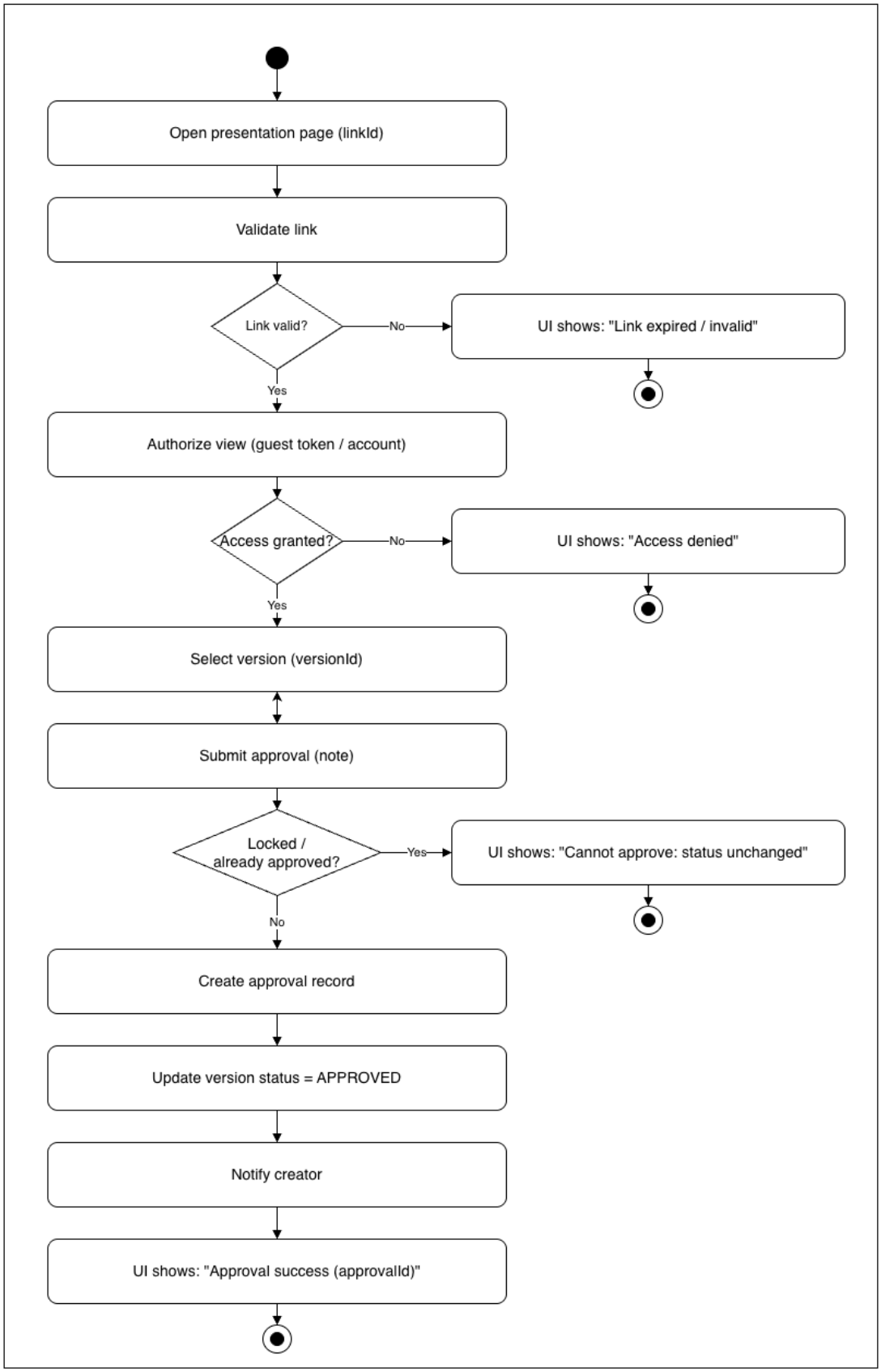
Activity Diagram: Approve Version